

Installing Docker Engine and Running WebGoat on Linux Distributions

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1 Objective

This guide provides step-by-step instructions for installing Docker Engine using the terminal on various Linux distributions including Debian and Ubuntu. It concludes with how to pull and run the WebGoat container and access it through a browser.

2 Prerequisites

- A Debian or Ubuntu-based Linux system
- Terminal access with `sudo` privileges
- Internet connection

3 Installing Docker Engine

3.1 For Debian 12+ (Bookworm)

1. Update and install required dependencies:

```
sudo apt update
sudo apt install apt-transport-https ca-certificates curl
software-properties-common gnupg lsb-release -y
```

2. Add Docker's official GPG key:

```
curl -fsSL https://download.docker.com/linux/debian/gpg |
sudo gpg --dearmor -o /usr/share/keyrings/docker.gpg
```

3. Add Docker's repository:

```
echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr
/share/keyrings/docker.gpg] \
https://download.docker.com/linux/debian $(lsb_release -cs)
stable" | \
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

4. Install Docker:

```
sudo apt update
sudo apt install docker-ce docker-ce-cli containerd.io
docker-compose-plugin -y
```

3.2 For Ubuntu 20.04+

1. Install required packages:

```
sudo apt update
sudo apt install apt-transport-https ca-certificates curl
software-properties-common gnupg lsb-release -y
```

2. Add Docker GPG key:

```
curl -fsSL https://download.docker.com/linux/ubuntu/gpg |
sudo gpg --dearmor -o /usr/share/keyrings/docker.gpg
```

3. Add Docker repository:

```
echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr
/share/keyrings/docker.gpg] \
https://download.docker.com/linux/ubuntu $(lsb_release -cs)
stable" | \
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

4. Install Docker Engine:

```
sudo apt update
sudo apt install docker-ce docker-ce-cli containerd.io
docker-compose-plugin -y
```

3.3 Verify Docker Installation

```
docker --version
```

4 Allow Docker to Run Without sudo

1. Add your user to the `docker` group:

```
sudo usermod -aG docker $USER
```

2. Apply group changes:

```
newgrp docker
```

Or log out and back in.

5 Running WebGoat in Docker

5.1 Pull WebGoat Image

```
docker pull webgoat/webgoat
```

5.2 Run the WebGoat Container

```
docker run -d -p 8080:8080 webgoat/webgoat
```

- `-d`: Runs in detached mode
- `-p 8080:8080`: Maps port 8080 on host to container

6 Accessing WebGoat via Browser

Open your web browser and navigate to:

`http://localhost:8080/WebGoat`

Default Login:

- Username: `guest`
- Password: `guest`

7 Why Use WebGoat?

WebGoat is a deliberately insecure web application maintained by OWASP to teach and test common web vulnerabilities such as:

- Broken Access Control
- SQL Injection
- XSS (Cross-Site Scripting)
- Insecure Deserialization

This makes WebGoat a valuable hands-on learning environment for students, ethical hackers, and security professionals.

8 Stopping or Removing Containers (Optional)

8.1 Stop a Container

```
docker ps
docker stop <container_id>
```

8.2 Remove a Container

```
docker rm <container_id>
```